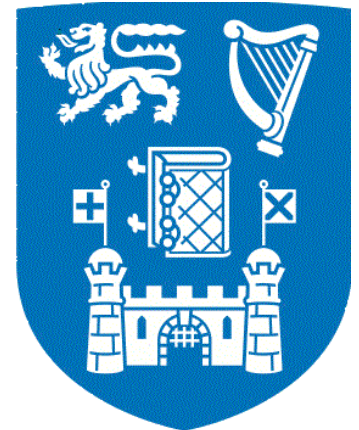


Formal Verification CSU44004/CSU55004

Lecture 3: Propositional Logic

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Recap

- ❖ Logic
- ❖ Propositional Logic
 - ❖ Symbolic Propositional Logic
 - ❖ Syntax
 - ❖ $A, C, B ::= p \mid (\neg A) \mid (A \wedge A) \mid (A \vee A) \mid (A \rightarrow A)$
 - ❖ Parse/Syntax Trees
 - ❖ Semantics
 - ❖ $\text{Sem}(p) \in \{T, F\}$
 - ❖ Example: $\text{Sem}(p) = T$, or $\text{Sem}(p) = F$
 - ❖ Formula and Model

Satisfiability - Validity

❖ Definition

- ❖ A formula (A) is *satisfiable* if there is at least one model that makes it true..
- ❖ A formula (A) is *falsifiable* if there is at least one model that makes it false..
- ❖ A formula (A) is *valid* (or a tautology) if it is true in every model..
- ❖ A formula (A) is a *contradiction/invalid* if it is false in every model..
- ❖ Q: which of the above are properties of
 - ❖ $(p \rightarrow \neg q) \rightarrow (q \vee \neg p)$
- ❖ Q: show that if $A1 \wedge (A2 \wedge A3)$ is satisfiable then $(A1 \wedge A2) \wedge A3$ is satisfiable.
- ❖ Q: Let $(A1 \wedge A2) \rightarrow A3$ be valid. Is it necessary that $A3$ is satisfiable?

Logic

❖ If the train arrives late and there are no taxi's at the station, then John is late for his meeting.

❖ John is not late for his meeting

❖ The train did arrive late

❖ Therefore?

❖ There were taxi's at the station

❖ p: "The train is late"

❖ q: "There are taxis in the station"

❖ r : "John is late for his meeting"

❖ $(p \wedge \neg q) \rightarrow r$

Lets prove our logic using contrapositive method

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$(p \wedge \neg q) \rightarrow r \equiv \neg r \rightarrow \neg (p \wedge \neg q) \text{ (contrapositive)}$$

$$= \neg (\neg r) \vee \neg (p \wedge \neg q) \quad [p \rightarrow q = \neg p \vee q]$$

$$= r \vee \neg p \vee q$$

$$= F \vee F \vee q$$

$$= F \vee q$$

$$= q$$

Reasoning

❖ **Q:** show that if $A1 \wedge (A2 \wedge A3)$ is satisfiable then $(A1 \wedge A2) \wedge A3$ is satisfiable.

❖ Suppose $A1 \wedge (A2 \wedge A3)$ is satisfiable.

❖ Then there is at least one assignment of truth values for which

$$A1 \wedge (A2 \wedge A3) = T$$

❖ Now evaluate

$$(A1 \wedge A2) \wedge A3$$

Reasoning

- ❖ **Q:** Let $(A1 \wedge A2) \rightarrow A3$ be valid. Is it necessary that $A3$ is satisfiable?
- ❖ A valid formula must be true for every truth assignment.
- ❖ Assume $A3$ is not satisfiable.
- ❖ Then $A3$ is false; $A3=F$.
- ❖ Consider the truth assignment
$$A1=T, A2=T, \text{ and } A1 \wedge A2=T.$$
- ❖ The implication becomes
$$T \rightarrow F = F$$
- ❖ This contradicts the assumption that $(A1 \wedge A2) \rightarrow A3$ is valid.
- ❖ Therefore, $A3$ cannot be always false.
- ❖ $(A1 \wedge A2) \rightarrow A3$ valid $\Rightarrow A3$ is satisfiable

Reasoning

- ❖ Our goal is to use the logic to derive logical conclusions from logical premises (assumptions).
- ❖ Two ways to do this:
 - ❖ **Syntactic reasoning**: using syntactic axioms and derivation rules. The preferred way because it's easier. We'll see this in the next lecture.
 - ❖ **Semantic reasoning**: using the truth values. We already did this in the associativity questions.

Semantic Reasoning

- ❖ Derive logical conclusions from the truth values.
- ❖ **Definition**
 - ❖ A set of premises $A1, A2, A3, \dots, An$ semantically entails a conclusion B , written as:
$$A1, A2, A3, \dots, An \models B,$$
to mean that any valuation giving to all $A1 \dots An$, the value **T** also gives B the value **T**. This is called *semantic entailment*.
 - ❖ We call $A1, \dots, An$ the *premises* or *antecedents*.
 - ❖ We call B the *conclusion* or *consequent*.
 - ❖ We *assume* that any valuation makes the empty premises **T**, meaning **B** is valid.
 - ❖ $\emptyset \models B$

Semantic Reasoning

❖ In summary:

- ❖ Premises **true**, conclusion **false** $\rightarrow$ *entailment fails*.
- ❖ Premises **false**, conclusion **true** $\rightarrow$ *allowed*.
- ❖ Premises **false**, conclusion **false** $\rightarrow$ **allowed**.
- ❖ Premises **true**, conclusion **true** $\rightarrow$ *required whenever the premises are true*.

Example: Club Membership

- ❖ Consider the claim: "*Everyone who is a club member gets a badge*".
 - ❖ $M = \text{"club member"}$
 - ❖ $B = \text{"gets a badge"}$
- ❖ The claim is: $M \models B$.
- ❖ Person 1
 - ❖ Club member: **Yes**, Badge: **No**.
 - ❖ $M=T, B=F$, *Entailment fails*.
- ❖ Person 2
 - ❖ Club member: **No**, Badge: **Yes**,
 - ❖ $M=F, B=T$, *Entailment allowed*
- ❖ Person 3
 - ❖ Club member: **No**, Badge: **No**,
 - ❖ $M=F, B=T$, *Entailment allowed*
- ❖ Person 4
 - ❖ Club member: **Yes**, Badge: **Yes**,
 - ❖ $M=T, B=T$, *Approved*

Semantic Reasoning

❖ **Example:** $P \wedge Q \models P$

P	Q	$P \wedge Q$	Conclusion P
T	T	T	T
T	F	F	T
F	T	F	F
F	F	F	F

❖ Therefore $P \wedge Q \models P$.

Semantic Entailment

- ❖ How to read *iff*?
- ❖ *"if and only if" or "both directions are true."*
- ❖ $A \Leftrightarrow B = (A \rightarrow B) \wedge (B \rightarrow A)$
- ❖ Prove the following using the semantics
- ❖ **Q:** Show that $p \rightarrow q \equiv \neg q \rightarrow \neg p$
- ❖ **Q:** Show that $\neg(p \wedge q) \equiv \neg p \vee \neg q$ and $\neg(p \vee q) \equiv \neg p \wedge \neg q$ (De Morgan laws)
- ❖ **Q:** show that $A \rightarrow B \equiv \neg A \vee B$.

Semantic Entailment

❖ Lemma

❖ $p \rightarrow q \equiv \neg q \rightarrow \neg p$.

❖ we have to prove two things:

❖ $p \rightarrow q \models \neg q \rightarrow \neg p$.

❖ $\neg q \rightarrow \neg p \models p \rightarrow q$

p	q	$\neg q \rightarrow \neg p$	$p \rightarrow q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

p	q	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

Indirect Proofs of Semantic Entailment

- ❖ Example: check if the following holds
 - ❖ $(p \rightarrow (q \vee r)), (q \rightarrow r), p \models (\neg r \rightarrow s)$
- ❖ There are 4 atomic propositions. How many lines in the truth table?
- ❖ $2^4 = 16$ lines!
- ❖ In general, the truth table necessary for checking $A_1, \dots, A_n \models B$ has 2^N lines, where N is the number of atomic propositions.
- ❖ $\rightarrow$ Brute force algorithm for checking validity is exponential.

Indirect Proofs of Semantic Entailment

- ❖ Indirect proof:
 - ❖ 1. assume entailment is falsifiable
 - ❖ 2. check if that's possible
 - ❖ 3. If you get a contradiction, no such valuation exists.
 - ❖ 4. Hence entailments holds.
- ❖ Falsifiable iff there is a valuation making all premises T and the conclusion F.
- ❖ Example: check if the following entailment holds:
 - ❖ $p \rightarrow q, p \models q$
 - ❖ 1. $p \rightarrow q = T, p = T, q = F$
 - ❖ 2. Since $p = T$ and $q = F, p \rightarrow q = F$
 - ❖ But we required $p \rightarrow q = T$. Contradiction.
 - ❖ 3. Contradiction: Therefore, the entailment holds.

indirect proofs of semantic entailment

- ❖ Example: Lets check the following

$$(\neg p \rightarrow q), (q \rightarrow p), (p \rightarrow \neg q) \models (p \wedge \neg q)$$

- ❖ Assume the entailment is falsifiable, this means

$$\neg p \rightarrow q = T$$

$$q \rightarrow p = T$$

$$p \rightarrow \neg q = T$$

$$p \wedge \neg q = F$$

- ❖ Analyze the premises
- ❖ Determine possible values
 - ❖ Case 1: $p=F$
 - ❖ Case 2: $p=T$

Premises 1: Using $A \rightarrow B \equiv \neg A \vee B$,
 $\neg p \rightarrow q \equiv p \vee q$.
So at least one of p, q must be true.

Premises 2: $q \rightarrow p$
This is false only when $q = T$ and $p = F$.
So, if q is true, p must also be true.

Premises 3: $p \rightarrow \neg q = T$
This is false only when $p = T$ and $q = T$.
Therefore, p and q cannot both be true.

p	q	$\neg p$	$\neg p \rightarrow q$
T	T	F	T
T	F	F	T
F	T	T	T
F	F	T	F

q	p	$q \rightarrow p$
T	T	T
T	F	F
F	T	T
F	F	T

p	q	$\neg q$	$p \rightarrow \neg q$
T	T	F	F
T	F	T	T
F	T	F	T
F	F	T	T

p	q	$\neg q$	$p \wedge \neg q$
T	T	F	F
T	F	T	T
F	T	F	F
F	F	T	F

